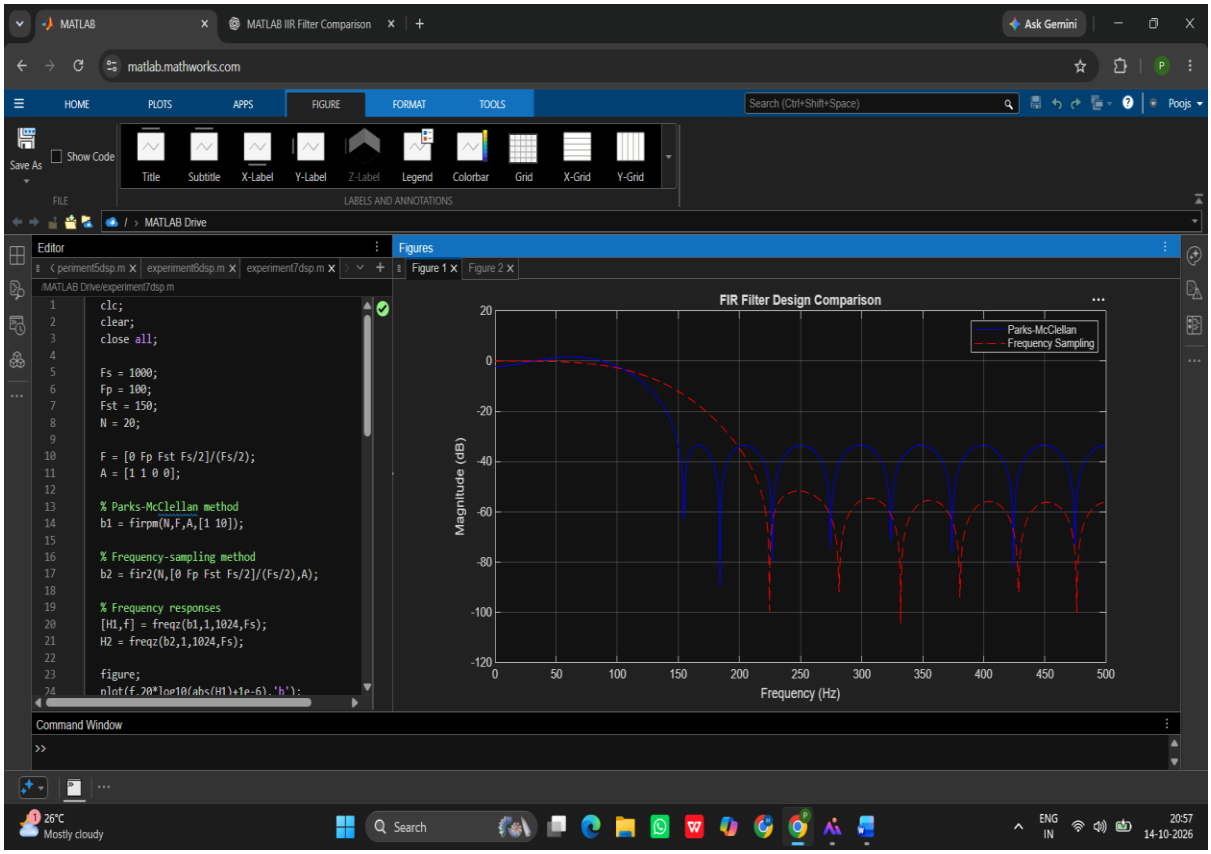
EXPERIMENT 7

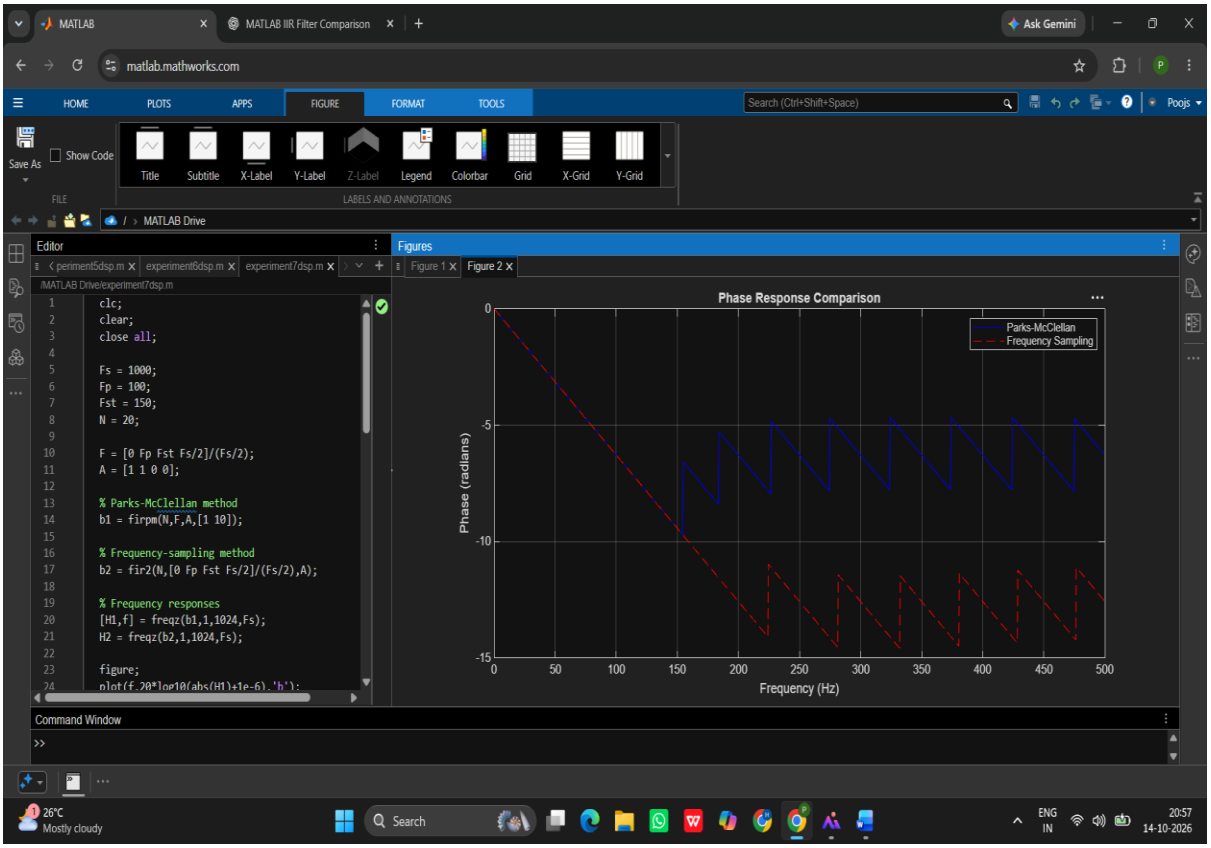
NAME: Pooja V

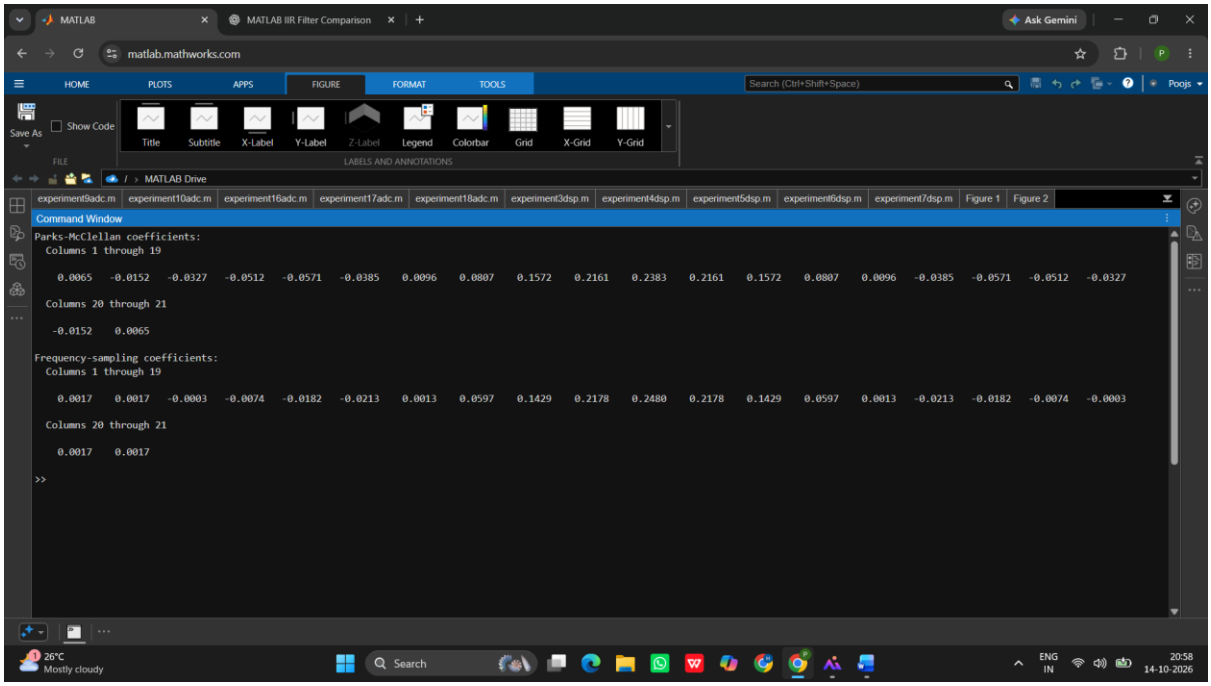
REG NO: 192512463

PROGRAM:

```
clc;
clear;
close all;
Fs = 1000;
Fp = 100;
Fst = 150;
N = 20;
F = [0 Fp Fst Fs/2]/(Fs/2);
...xlabel('Frequency (Hz)');
ylabel('Phase (radians)');
title('Phase Response Comparison');
legend('Parks-McClellan','Frequency Sampling');
disp('Parks-McClellan coefficients:');
disp(b1);
disp('Frequency-sampling coefficients:');
disp(b2);
```


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periment5dsp.m
experiment6dsp.m
experiment7dsp.m
/MATLAB Drive/experiment7dsp.m
clc;
clear;
close all;
Fs = 1000;
Fp = 100;
Fst = 150;
N = 20;
F = [0 Fp Fst Fs/2]/(Fs/2);
A = [1 1 0 0];
% Parks-McClellan method
b1 = firpm(N,F,A,[1 10]);
% Frequency-sampling method
b2 = fir2(N,[0 Fp Fst Fs/2]/(Fs/2),A);
% Frequency responses
[H1,f] = freqz(b1,1,1024,Fs);
H2 = freqz(b2,1,1024,Fs);
figure;
plot(f,20*log10(abs(H1)+1e-6),'b');
Figures
Figure 1
Figure 2
FIR Filter Design Comparison
Parks-McClellan
Frequency Sampling
Magnitude (dB)
Frequency (Hz)
20
0
-20
-40
-60
-80
-100
-120
0
50
100
150
200
250
300
350
400
450
500
Command Window
>>
26°C
Mostly cloudy
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Y-Grid
FILE
LABELS AND ANNOTATIONS
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Figures
Figure 1
Figure 2
/MATLAB Drive/experiment7dsp.m
clc;
clear;
close all;
Fs = 1000;
Fp = 100;
Fst = 150;
N = 20;
F = [0 Fp Fst Fs/2]/(Fs/2);
A = [1 1 0 0];
% Parks-McClellan method
b1 = firpm(N,F,A,[1 10]);
% Frequency-sampling method
b2 = fir2(N,[0 Fp Fst Fs/2]/(Fs/2),A);
% Frequency responses
[H1,f] = freqz(b1,1,1024,Fs);
H2 = freqz(b2,1,1024,Fs);
figure;
Phase Response Comparison
Parks-McClellan
Frequency Sampling
Phase (radians)
Frequency (Hz)
0
-5
-10
-15
0
50
100
150
200
250
300
350
400
450
500
Command Window
26°C
Mostly cloudy
Search
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IN
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Z-Label
Legend
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experiment17adc.m
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experiment4dsp.m
experiment5dsp.m
experiment6dsp.m
experiment7dsp.m
Figure 1
Figure 2
Command Window
Parks-McClellan coefficients:
Columns 1 through 19
0.0065 -0.0152 -0.0327 -0.0512 -0.0571 -0.0385 0.0096 0.0807 0.1572 0.2161 0.2383 0.2161 0.1572 0.0807 0.0096 -0.0385 -0.0571 -0.0512 -0.0327
Columns 20 through 21
-0.0152 0.0065
Frequency-sampling coefficients:
Columns 1 through 19
0.0017 0.0017 -0.0003 -0.0074 -0.0182 -0.0213 0.0013 0.0597 0.1429 0.2178 0.2480 0.2178 0.1429 0.0597 0.0013 -0.0213 -0.0182 -0.0074 -0.0003
Columns 20 through 21
0.0017 0.0017
>>
26°C
Mostly cloudy
Search
ENG
IN
20:58
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